Project Consulting Report

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1. Executive Summary

The project aims to deliver a multi?tenant, white?label SaaS platform that provides AI?powered conversational role?play training for retail sales associates. By leveraging Azure serverless services, OpenAI language models, and 11?Labs voice synthesis, the solution will enable scalable, real?time coaching, automated scoring, and reporting that links practice activity to sales performance. Key decisions include an Azure?centric architecture with React front?end, Azure Functions backend, and Cosmos DB for multi?tenant data isolation, all within a constrained MVP budget of $500?800?K and a tight pilot timeline. The recommendation is to proceed with the proposed architecture, focusing on rapid MVP delivery for 73 pilot stores, followed by phased enhancements and cloud?agnostic extensions.

2. Background Summary

The client currently relies on limited in?person training, where a single trainer supports 135 stores, resulting in inconsistent role?playing experiences and insufficient coaching capacity. Associates have only brief daily windows for practice, and existing off?the?shelf tools fail to meet defined KPIs, lacking integration with POS data and white?label capabilities.

3. Problem / Need Analysis

Problem / Need	Business Impact
Current training is limited because only one person can train 135 st	Inconsistent training reduces associate skill development, leading to lower
Many stores lack expert staff to conduct role?play sessions, creating coaching gaps	Coaching gaps increase turnover and diminish customer experience qualit
Sales associates have only 3?5 minutes per day for practice due to	Limited practice time hampers skill reinforcement, resulting in sub?optimal
Off?the?shelf tools do not meet all defined KPIs and require workarounds.	Workarounds increase operational overhead and prevent accurate measur
There is no integration with POS systems, making it difficult to link t	Inability to correlate training with sales data obscures ROI and makes it ha
Rapid evolution of AI technology creates uncertainty about future support and maintenance	Future?proofing risk may lead to costly re?engineering or loss of competiti
The client needs to keep development costs low while delivering a	Budget overruns could jeopardize project approval and delay time?to?valu
Uncertainty about the client?s cloud environment (Azure vs AWS) adds planning complexity.	Potential cloud?vendor lock?in could limit scalability and increase migratio
Manual data comparison is required for effectiveness metrics due t	Manual effort reduces insight speed and increases risk of inaccurate repor
Without a white?label solution, duplicating the tool across multiple p	Inefficient duplication raises operational costs and limits market expansion

4. Requirements Analysis

Functional Requirements

ID	Requirement
FR-001	The platform must provide an AI?powered conversational interface for sales role?play tr...
FR-002	It must allow upload and management of training content such as brochures, PowerPoint d...
FR-003	The solution should support voice interaction using OpenAI for language modeling and 11...
FR-004	It must generate real?time scoring and feedback during role?play sessions.
FR-005	Each role?play interaction must be recorded and stored as a transcript for manager review.
FR-006	The system should track practice frequency and score improvement for each associate.
FR-007	It must provide reporting that links training activity to attach?rate uplift in sales d...
FR-008	Managers need the ability to add notes, adjust scoring, and fine?tune AI recommendations.
FR-009	The platform should allow configuration of difficulty levels and objection?handling sce...
FR-010	It must be delivered as a white?label SaaS solution that can be duplicated across multi...
FR-011	The MVP should support multi?tenant access for approximately 73 pilot stores initially.
FR-012	Manual comparison metrics are required for the MVP without POS integration.
FR-013	The solution must enable export or viewing of reports at both individual associate and ...

Non-Functional Requirements

Category	Requirement
Performance	User interfaces and conversational turns shall respond within 2 seconds under normal load.
Security	All data at rest and in transit must be encrypted; strict tenant isolation enforced; co...
Scalability	Platform must scale to support 100+ stores and additional tenants without performance d...
Availability	Target 99.9% uptime using Azure managed services and automated failover.
Maintainability	Codebase modularized with CI/CD pipelines to allow automated testing, deployments, and ...

Constraints

Constraint
The MVP budget is limited to $500?800?K.
A pilot must be ready before October, with test stores active by the end of June.
No POS integration is planned for the initial proof of concept.
The client expects proposal submissions by the end of next week and a decision after in...
The client requires ownership of the IP so the solution can be white?labelled and marke...
The client?s cloud provider has not been confirme

Business Goals

Goal

Technology Context

Technology Context
Preferred Cloud: Azure
Existing CRM: Salesforce

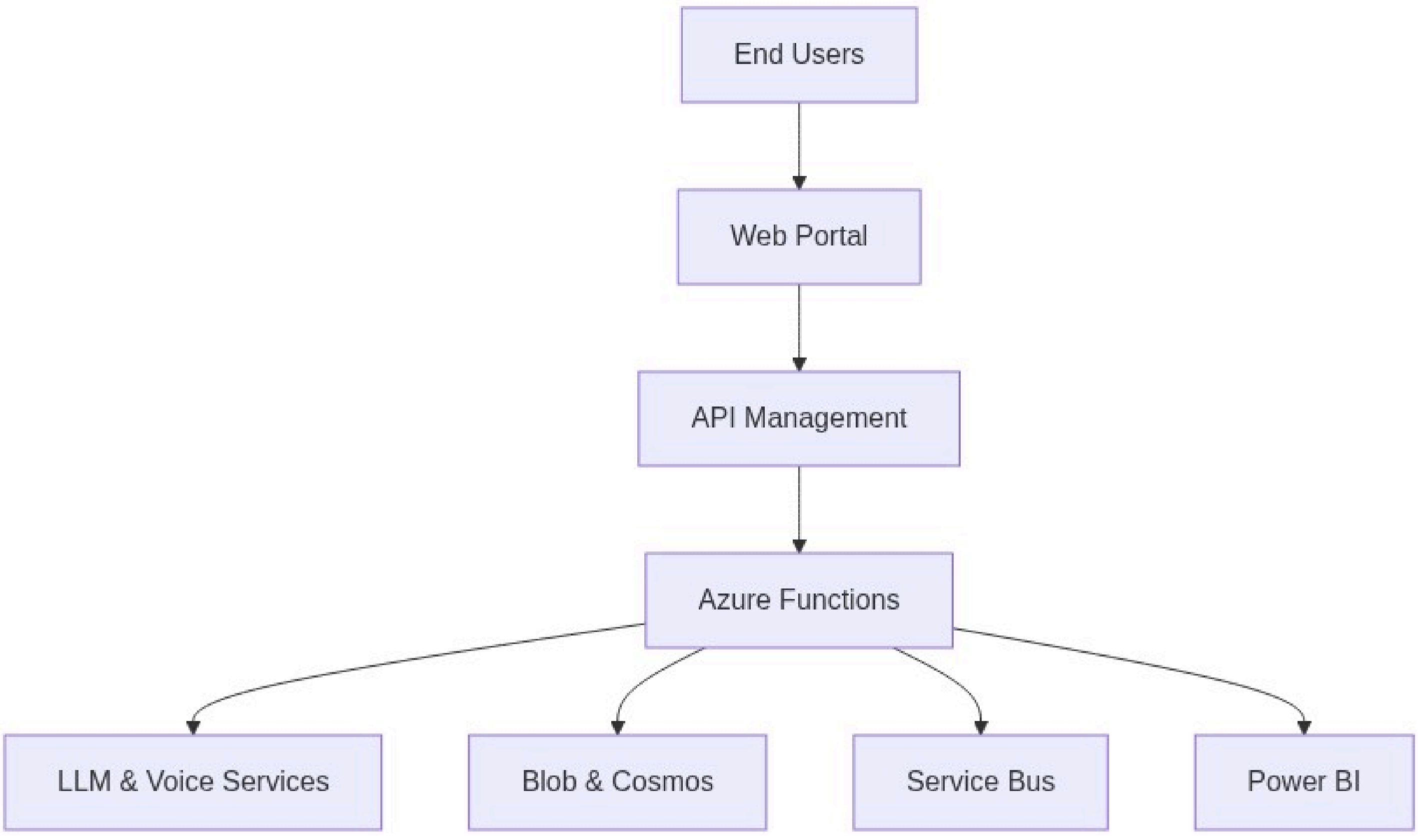
5. Assumptions

Assumption
Azure subscription is available and provisioned.
Client will provide API credentials for third?party integrations.
Existing infrastructure remains unchanged during the POC phase.

6. Feature & Module Breakdown

Module	Feature / Functionality	Technologies Used
Frontend	Responsive UI for scenario selection, real?time interaction, dashbo	React 18 hosted on Azure Static Web App
Backend API	REST/GraphQL endpoints for content management, scoring, transc	Azure Functions (Node.js 20)
Authentication	Multi?tenant user management, JWT issuance, password reset.	Azure AD B2C, Azure Key Vault
API Gateway	Routing, throttling, tenant isolation policies.	Azure API Management (Developer tier)
Storage	Persistent storage for PDFs, PPTX, audio recordings, transcripts.	Azure Blob Storage (Cool tier)
Database	Multi?tenant user profiles, scores, notes, session metadata.	Azure Cosmos DB (Core (SQL) API)
Search	Semantic search over uploaded training content for contextual AI re	Azure Cognitive Search
Message Queue	Asynchronous processing of content ingestion, scoring pipelines, a	Azure Service Bus (Standard tier)
CI/CD	Automated build, test, and deployment for both Azure and potential	Azure DevOps Pipelines (YAML)
Monitoring	Telemetry, health checks, usage analytics, alerting.	Azure Monitor + Application Insights
Reporting	Manager dashboards, exportable CSV/Excel reports at associate a	Power BI Embedded (shared capacity)
Voice Synthesis Integration	Generate realistic voice responses for role?play scenarios.	11?Labs Voice Generation API (HTTPS ca
Speech?to?Text	Transcribe associate speech to text for scoring and storage.	Azure Speech Service (Speech?to?Text)

7. Solution Architecture



8. Feasibility Assessment

Metric	Value	Reason
Complexity	High	Multi?tenant architecture, AI integration, and tight timeline increase development
Architecture Confidence	High	Proven Azure services and design alignment with client constraints give strong co
Feasibility Confidence	Medium	Budget and timeline constraints require careful cost monitoring and performance

9. Risk Assessment

Risk	Impact	Mitigation Strategy
Cost overruns from Cosmos DB provisioned t	Budget could exceed the $5	
High latency in the voice pipeline (speech?to?text ? LLM ? text?to?speech) causes poor	Associates may abandon practice sessions, reducing training effectiveness and ROI metrics.	
Insufficient tenant data isolation in Cosmos DI	Compliance breach (ISO?27	...
Dependency on Azure OpenAI service; if the client later selects AWS,	Potential vendor lock?in, and increased operational complexity for a multi?cloud rollout.	

10. Recommendations & Next Steps

Begin immediate onboarding of a cross?functional delivery team comprising a solution architect, senior developers (frontend and backend), DevOps engineer, and QA lead. Prioritize provisioning of Azure resources (AD B2C, Cosmos DB, API Management) and setting up CI/CD pipelines within the next two weeks. Deliver the MVP core flow?authentication, content upload, voice?enabled role?play, real?time scoring, and transcript storage?by the end of June to meet the pilot rollout deadline. Conduct weekly cost?monitoring reviews and performance tests to stay within the $500?800?K budget and latency targets. After the pilot, plan phased enhancements: POS data integration, expanded tenant isolation, and migration to AWS using the existing IaC abstractions.

11. Architecture Summary

The Azure?centric, serverless architecture was chosen because it offers built?in AI services, multi?tenant data isolation, and pay?as?you?go pricing that align with the client?s low?budget, fast?time?to?market goals. React on Azure Static Web Apps provides a lightweight, cost?effective UI, while Azure Functions deliver scalable compute for conversational flows and scoring. Cosmos DB and Azure Blob Storage ensure secure, isolated storage for each tenant?s data, and API Management provides a single, governed entry point essential for a white?label SaaS offering. Together these components satisfy functional and non?functional requirements while remaining flexible for future cloud?agnostic expansion.